

Exposé on Master's Thesis

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1 Topic

Threading a nut onto a bolt is one of the most common operations in industrial assembly, yet it remains hard to automate: the parts mate through sub-millimetre thread geometry, progress is governed by friction and contact forces that are difficult to sense, and success couples a stable grasp of the fastener to a controlled turning motion of the grasped part. A system meant to perform this task without part-specific fixturing must therefore acquire and hold the grasp itself, sense contact through that grasp, and coordinate grasping with fine, rotation-coupled manipulation.

Reinforcement learning (RL) in high-fidelity GPU simulation has recently made contact-rich assembly learnable, and nut threading has been demonstrated among such tasks [1, 2]. These learned systems, however, act with a parallel-jaw gripper and begin each episode from a fastener that is already held: the policy only ever learns to turn a pre-pinch part and never has to acquire, or actively maintain, the grasp itself. Grasp acquisition and maintenance are thus left outside the learning problem.

Multi-fingered dexterous hands, by contrast, promise far more general manipulation. Learned skills now span in-hand reorientation [3, 4], tool use [5], and bimanual tasks such as twisting lids off jars [6], and robotic fastening has been studied with guided screwdrivers [7, 8]. Yet dexterous hands remain markedly harder to control than grippers, precisely because of the many degrees of freedom and the rich, intermittent, hard-to-sense finger–object contact they rely on [9, 10]. The intersection of the two – a multi-fingered hand that performs a contact-rich fastening task, forming the grasp and then executing the turn – is therefore largely unexplored: to the author's knowledge, no learned policy both acquires a multi-finger grasp of a fastener and performs the subsequent threading on an arm–hand system.

This thesis aims to close part of that gap. It pursues a two-stage approach that stays deliberately general about the concrete hardware. First, a privileged teacher policy is trained in simulation to reach for a nut resting on a bolt, pinch it with a multi-fingered hand, and thread it onto the bolt. This teacher is then distilled into a vision-based student policy that replaces the privileged simulator state with RGB – and, where needed, RGB-D – observations, and is thus in principle transferable to real hardware – following the teacher–student recipe the author validated for dexterous grasping in a preceding semester thesis [11]. Relative to gripper-based assembly the task adds a reach-and-grasp phase, a side-pinch of a low-friction fastener, and the requirement to turn it down the thread into a tight success window without losing the grip. The relevance is threefold. Methodologically, the

task probes credit assignment across a coupled grasp-then-manipulate sequence, contact sensing on a dexterous hand, and the limits of vision-based distillation for fine manipulation. From a simulation standpoint, it stresses how faithfully GPU physics engines represent mating thread contact – penetration-free contact simulation being an open research problem in its own right [12]. Practically, it is a step toward replacing special-purpose grippers with general dexterous hands for fastening, and, in the longer term, toward transferring such skills from simulation to reality [4, 13].

1.1 Research questions

The thesis pursues five research questions:

1. Can a single PPO policy learn the coupled grasp-then-thread skill on a humanoid arm–hand system in simulation, starting without a pre-formed grasp?
2. Which contact-modelling and contact-sensing choices make the task representable and learnable in the first place – from the physics backend’s representation of the mating thread geometry to filtered finger–nut contact signals in place of geometric proxies?
3. How must the reward be structured so that the policy performs genuine rotation-coupled threading instead of exploiting geometric progress signals (align-and-hold, pressing to depth without rotation)?
4. How well does the learned skill generalize across nut/bolt sizes (e.g. M10 versus M16), and how far can threading depth and multi-turn rotation be pushed?
5. Can the privileged teacher be distilled into a vision-based student policy operating on stereo RGB or RGB-D observations, and which observation design – resolution, depth information, egocentric versus third-person camera placement – provides the object localization that threading requires, as the basis for sim-to-real transfer?

2 Approach

The approach builds on a vendored copy of the IsaacLab *forge* task [2] – itself built on *factory* [1] – and carries it from the Franka parallel-jaw gripper to the KUKA–SHARPA system. Figure 1 sketches the setup. The following subsections describe the platform and robot, the task and its control interfaces, the reward and contact design, the learning method, and the distillation to a vision-based student policy.

2.1 Simulation platform and robot

Training runs in IsaacLab on Isaac Sim, which simulates contact-rich assembly across many environments in parallel on the GPU. Development started on IsaacLab v2.3.2 with the PhysX backend; the task has since been migrated to the Newton/MJWarp backend of IsaacLab 3.0. The migration was driven by contact fidelity: under PhysX, the nut, modelled with a signed-distance-field (SDF) collider, could be pressed axially through the bolt thread without rotating – an exploit the policy reliably discovered – while genuinely meshing thread contact was not available as a learning signal. Newton’s collision pipeline supports mesh–mesh SDF colliders and treats the nut–bolt pair with a hydroelastic pressure-field contact model, so the mating thread geometry is physically represented. Complementary to this high-fidelity line, a second pipeline is planned that approximates the thread mating as a kinematic helical constraint (in the spirit of the vendored SimToolReal task line [5]), trading contact

fidelity for simulation speed; the two pipelines are developed in parallel and compared. The robot is a KUKA iiwa14 (7-DoF) arm with a left SHARPA hand (22 finger joints). Only the thumb (5) and index (4) are active, driven through a low-dimensional pinch synergy; the middle, ring, and little fingers are frozen curled away to remove irrelevant degrees of freedom and self-collision clutter. The task is realized as a new `forge_kuka` task family with its own gym id, leaving the original Franka forge tasks untouched.

2.2 Task and control

The nut sits physically on the bolt and is laterally constrained by the shaft, so the policy must grip it in order to turn it. The 7-DoF arm is driven by a task-space impedance controller (operational-space control, $\tau = J^T(\Lambda \dot{w}^* + \mu + p)$), while the fingers are position-controlled. A *static, flange-anchored* end-effector control frame decouples arm control from finger motion, so gripping does not disturb the arm Jacobian; introducing this frame was what first made a firm pinch learnable. The base action space comprises end-effector pose increments (position and orientation) plus the pinch finger degrees of freedom; the yaw action is mapped unidirectionally to the tightening direction. In addition, the thesis investigates an object-centric action interface: a contact-level impedance controller through which the policy commands increments of the nut itself – rotation about the bolt axis, insertion depth, and grip force – and which converts these into fingertip targets. Such object-centric interfaces have proven effective for dexterous manipulation of grasped objects [5, 14].

2.3 Reward design and contact sensing

The reward combines `forge`’s multi-scale keypoint threading reward with a staged grasp shaping – approach → contact → perpendicular side-pinch → turn – in the spirit of composing grasping with in-hand manipulation [15, 16]; force closure of the pinch provides the classical background [17]. Two lessons from the experiments so far shape this design. First, the grasp and engagement terms must be gated on a *real, filtered* finger–nut contact signal: proxy signals – distance to the nut centre, unfiltered net contact forces – either mis-matched the geometry or were corrupted by finger–finger self-contact, and the breakthrough was to read the filtered pairwise contact force (`force_matrix_w`) of the finger pads against the nut’s *rigid-body link*, which separates the true grip from self-contact. Second, purely geometric progress and success signals are exploitable: trained policies learned to align and hold the nut above the bolt, to press it to depth without rotating, or to oscillate the wrist to farm the turning reward – all without threading. Success is therefore verified on rendered replays rather than on the environment’s geometric success signal alone, and a central thrust of the thesis is a rotation-coupled descent reward that only pays out when the nut simultaneously rotates and descends according to the thread’s pitch.

2.4 Learning algorithm and methodology

Learning uses PPO (`rl_games`) in an asymmetric actor-critic setup with a privileged critic state and an LSTM, under `forge`-style domain randomization (control gains, thresholds, action smoothing, friction). An accompanying method study compares vanilla PPO with the mixed-exploration SAPO variant for training the teacher. Development follows a strict *experiment cycle*: one configuration change per run, one commit per run, results recorded before committing, so that `git checkout` of a run reproduces its documented result. This places the work within the broader line of RL for robotics [13, 18] and, specifically, scaled dexterous arm–hand learning [19–21].

2.5 Distillation to a vision-based student policy

While the privileged teacher may exploit simulator-internal state (the exact nut pose, contact forces), the policy that is ultimately deployed can only rely on real sensors. The thesis therefore distills the teacher into a vision-based student that replaces the privileged object state with stereo RGB – and, if needed, RGB-D – observations plus proprioception. The pipeline follows the author’s semester thesis [11], which distilled a privileged grasping teacher for a 23-DoF Agile Hand on a Franka Research 3 into a stereo-RGB student (a ResNet encoder per view, cross-attention stereo fusion, an LSTM, and an auxiliary object-position loss, after DextrAH-RGB [21]): safety-aware SafeDagger distillation [22] retained about 98% of the teacher’s grasp success there (97.0% versus 99.0%) and dominated behavioural cloning and DAgger [23] on both task success and safety; the DexSafeDagger framework [24] continues this line with value-based intervention gating. Whether this recipe carries over to threading is open in one decisive respect: in the semester thesis the stereo student localized objects only to about 1–2 cm at 320×240 pixels, whereas threading demands centring within 2.5 mm. Which observation design closes this gap – higher resolution, explicit depth (RGB-D), egocentric versus third-person camera placement, additional auxiliary supervision – is a central experimental question of the thesis.

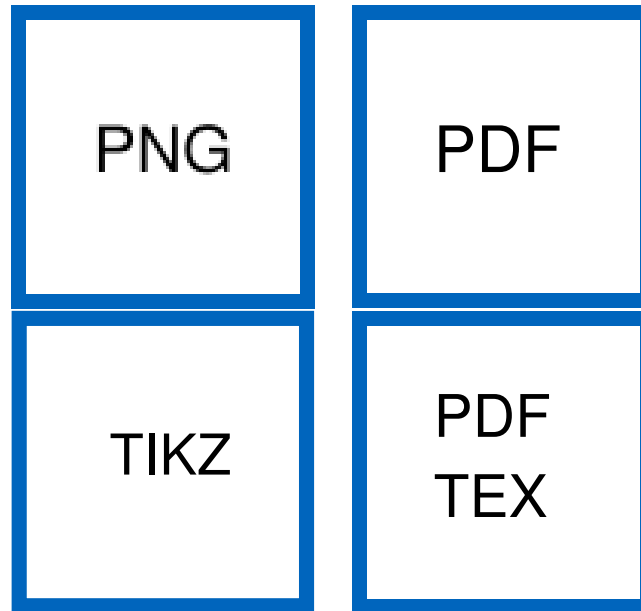


Figure 1: System overview (placeholder figures): the KUKA iiwa14 with SHARPA hand pinches an M16 nut with a thumb–index grip and threads it onto the standing bolt, driven by the staged grasp-then-thread reward.

3 Work Plan and necessary Resources

3.1 Methodology and evaluation

The primary success metric is threading success rate (nut centred, seated to target depth, correctly turned), with every success claim verified on rendered replays of the trained policy, since the environment’s geometric success signal has proven exploitable. Secondary metrics are net accumulated nut rotation and threading depth (nut–bolt distance), grip quality – contact rate, grip perpendicularity, contact force, and sustained-contact duration – and episode length. For the distilled student, the retained fraction of teacher performance and the object-localization accuracy achievable from the

visual observations (resolution, stereo versus RGB-D, camera viewpoint) are additionally evaluated. Because every configuration change is isolated in exactly one reproducible run, the contribution of each design decision (contact signal, grip gate, spawn pose, end-effector frame, physics backend, reward weights) can be established by ablation. Comparisons are made (i) against the pre-grasped forge baseline, (ii) between the two simulation pipelines, and (iii) across nut/bolt sizes, for which custom M10 assets with SDF colliders were authored. In the final experimental phase, a sim-to-real transfer of the student policy with hardware experiments on the physical KUKA–SHARPA system is planned, contingent on hardware availability.

3.2 Resources

The work requires: Isaac Sim with IsaacLab in uv-managed environments (v2.3.2/PhysX for the baseline line, 3.0 with the Newton/MJWarp backend for the main line); a multi-GPU workstation – contact-rich training runs take hours to days, and the hydroelastic SDF contact of the threading pair is computationally heavy – with browser-based remote inspection of running policies; the USD assets of the KUKA iiwa14 and the SHARPA hand, and the `forgeUltra` codebase. The sim-to-real part additionally requires a physical KUKA iiwa14 with a SHARPA hand and corresponding supervision.

3.3 Schedule

Figure 2 shows the planned sequence in five work packages: research and baseline experiments through mid-June (WP1), one month for implementing the teacher policy via two parallel simulation pipelines and a unified pick-place-screw agent through mid-July (WP2), student distillation and transfer to our own assets through mid-August (WP3), hardware transfer and experimentation through the first week of September (WP4), and finally further experiments, documentation, and writing through mid-November (WP5). Hands-on work began in June 2026, one month ahead of the official thesis start on 1 July (dashed line); an optional paper track (orange) bundles the final week into writing and further experiments, targeting an ICRA 2026 submission on 15 September.

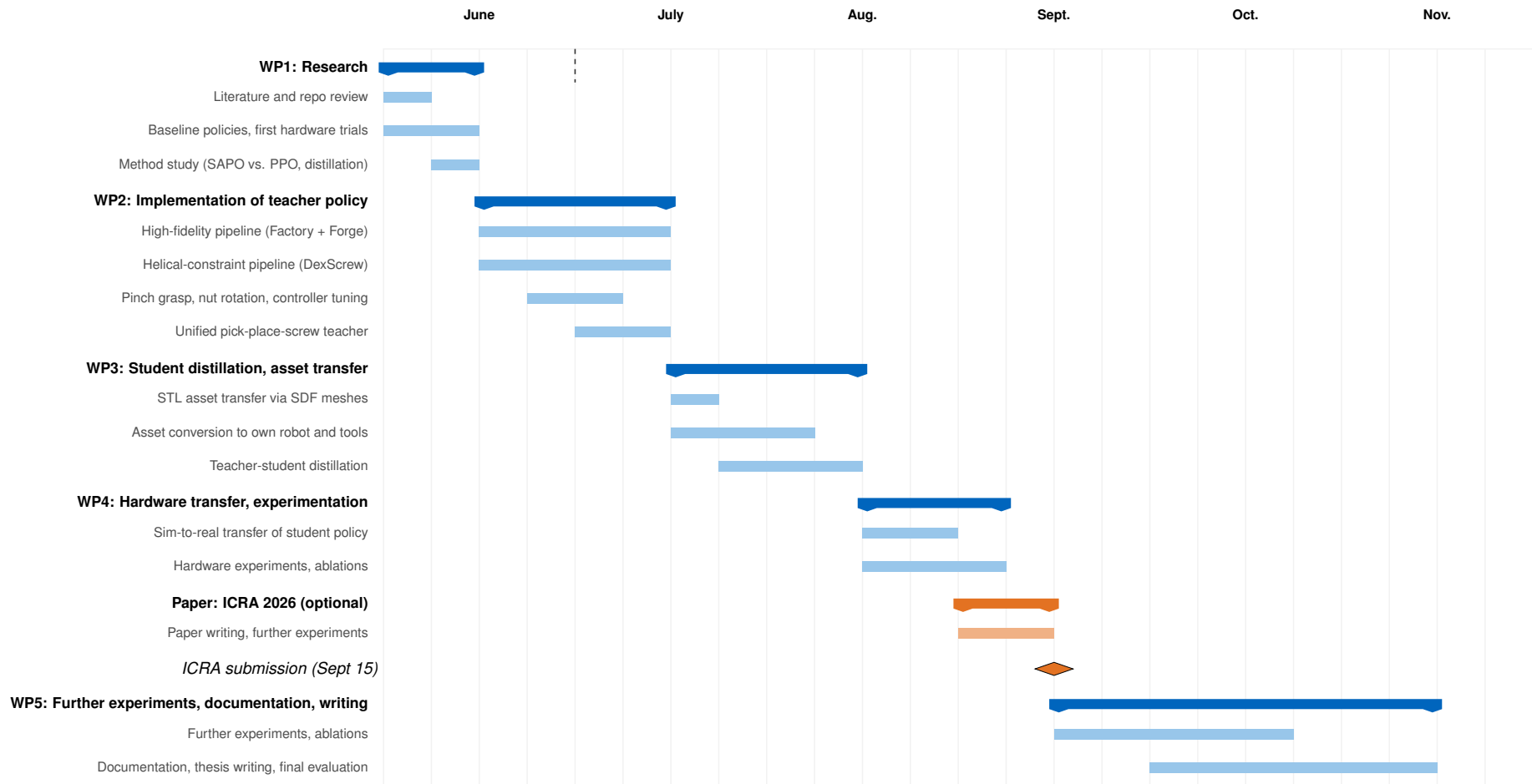


Figure 2: Time schedule (2026). The optional ICRA paper track is highlighted in orange; the dashed line marks the official thesis start (1 July).

4 Literature

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